Quantum Formulation of the Three-Body Problem in the OXI Framework

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Abstract

This paper presents a novel approach to the classical three-body problem using a higher-dimensional geometric framework known as the OXI (Orbital Cross-Interaction) formalism. By interpreting the three-body system as a "world piece computer" where each body performs its own physics computation, we unify concepts from quantum mechanics and celestial mechanics. We introduce a formalism based on discrete residue updates and quantum-like state transitions that emerge naturally from projections of higher-dimensional rotations. This approach provides new insights into the apparent chaos of the three-body problem as a projection artifact from a well-ordered system in seven-dimensional space (R^7).

1 Introduction

The three-body problem stands as one of the most challenging problems in classical mechanics, exhibiting chaotic behavior that has resisted complete analytical solutions for centuries. In this paper, we propose a novel perspective: viewing the three-body system as a "world piece computer" where each body actively computes its own trajectory by exchanging geometric information with other bodies.

Our approach builds upon the OXI (Orbital Cross-Interaction) framework, which posits that apparent complexity emerges from simple higher-dimensional geometry through projection. The core insight is that each physical entity actively computes its own trajectory by exchanging geometric information with other entities, following geodesics in a seven-dimensional space. The chaos we observe in three dimensions is interpreted as an artifact of projection from this higher-dimensional space.

The three-body system in classical mechanics consists of three masses interacting through gravitational forces. When viewed from our 3D perspective, the system exhibits chaotic behavior that makes long-term prediction challenging. However, by reformulating the problem in a higher-dimensional space (specifically a 7D space comprising 3 spatial dimensions, 3 time dimensions, and 1

master time dimension), we can reveal underlying regularities and symmetries not apparent in the conventional approach.

We will develop a quantum-inspired formalism for the three-body problem, introducing discrete "quantum" updates of the system state that correspond to accumulations of physical action. This approach creates a bridge between classical celestial mechanics and quantum-mechanical concepts, potentially providing new tools for understanding the long-term evolution of gravitationally bound systems.

The quantum-mechanical inspiration comes from recognizing parallels between our discretized approach and quantum state transitions. Just as electron orbitals have quantized angular momentum, we propose that the binary pairs in a three-body system can be understood as having discretized states that transition according to well-defined rules of action accumulation.

2 Discrete Residue Updates

For rapid dynamics, we can break down the full-cycle residue into discrete "quantum" updates. This creates a discretized mechanism where updates occur in well-defined steps rather than continuously.

2.1 Mathematical Formulation

The discrete residue update can be formalized as:

$$\Delta R_{\text{discrete}} = \frac{R}{n} \cdot \left\lfloor \frac{\Delta R_{\text{accumulated}}}{R/n} \right\rfloor \tag{1}$$

Where:

- R is one complete cycle residue (e.g., 2π for angular variables)
- n is the discretization factor (an integer determining update granularity)
- $\Delta R_{\text{accumulated}}$ is the continuously accumulated residue
- $\lfloor\rfloor$ represents the floor function, triggering updates only when sufficient residue has accumulated

This creates a discrete "ratcheting" mechanism where updates occur in well-defined steps rather than continuously, similar to quantum state transitions.

2.2 Update Thresholds

We can implement update thresholds using indicator functions:

$$I_{\text{update}} = \Theta\left(\max\left\{\frac{|\Delta E|}{E_{\text{ref}}} - \epsilon_E, \frac{|\Delta L|}{L_{\text{ref}}} - \epsilon_L, \frac{|\Delta r|}{r_{\text{ref}}} - \epsilon_r, \frac{|\Delta\varphi|}{2\pi/n} - 1\right\}\right) \tag{2}$$

Where:

- $\Theta(x)$ is the Heaviside step function (0 for $x < 0$, 1 for $x \geq 0$)
- ϵ values are the threshold constants
- Reference values (E_{ref}, L_{ref}, r_{ref}) normalize the changes
- $\Delta\varphi$ is the phase accumulation with discretization $2\pi/n$

When $I_{\text{update}} = 1$, the system applies the accumulated residues and resets the accumulators.

3 The "World Piece Computer" Perspective

The "world piece computer" perspective inverts the traditional approach to physics: rather than imposing differential equations on passive objects, we recognize that each physical entity actively computes its own trajectory by exchanging geometric information with other entities.

3.1 Distributed Computation Framework

In this view:

- Each body maintains its own local "register" of accumulated residues
- When bodies interact, they exchange information that updates these registers
- The actual physics emerges from each body following its geodesic in 7D space
- The chaos we observe is an artifact of projection to lower dimensions

The three-body problem becomes chaotic in 3D or 4D precisely because we've lost access to the full 7D information. In the complete 7D picture, all trajectories are indeed "straight lines" (geodesics) in the proper geometric context.

3.2 The Trivector Representation

A key mathematical object in our framework is the trivector Ω_{123} which represents the state of the three-body system in the 7D space. Its evolution is governed by:

$$\Omega_{123}(\tau) \leftrightarrow \Phi(\tau; q_x, t_x, q_y, t_y, q_z, t_z) \tag{3}$$

Where Φ represents the state function in the 7D space with coordinates $(q_x, t_x, q_y, t_y, q_z, t_z, \tau)$. The evolution of this function under the R_7^{OV} matrix corresponds to the real-valued equation:

$$\frac{\partial \Phi}{\partial \tau} = \hat{H}_7 \Phi \tag{4}$$

Where $\hat{H}_7$ is the 7D Hamiltonian operator containing all binary pair interactions. This approach aligns with both Wheeler's "it from bit" concept and modern interpretations of quantum mechanics as information exchange.

4 Quantum-Mechanical Approach to the Three-Body Problem

The discrete nature of our residue updates suggests a parallel with quantum mechanics. Just as electron orbitals have quantized angular momentum, we can treat the binary pairs in a three-body system as having discretized states.

4.1 Quantum-Inspired Formalism

We can define analogous quantization conditions for each binary pair in our three-body system:

$$J_{ij} = n_{ij} \cdot \hbar_{\text{eff}} \tag{5}$$

Where:

- J_{ij} is the action variable for binary pair (i, j)
- n_{ij} is an integer quantum number
- $\hbar_{\text{eff}}$ is an effective "Planck constant" for the system

This connects to Bohr's original quantization condition ($mvr = n\hbar$) but extends it to the three-body context where each binary pair has its own quantized angular momentum.

The "allowed" configurations would be those where all three binary pairs satisfy their quantization conditions simultaneously, and transitions between these states would occur through discrete residue updates.

This approach provides several advantages:

- Discretized Angular Momentum: Binary pairs' angular momenta exist only in discrete "allowed" states
- State Transitions: The system "jumps" between allowed configurations when sufficient residue accumulates
- Probability Amplitudes: We can work with probability distributions over possible configurations
- Simplified Computation: We can focus on discrete state variables rather than exact positions and velocities

5 Geometry of Action Quanta

To formalize the connection between quantum parameters and orbital geometry, we introduce a geometric interpretation of action quanta.

5.1 Action Shells and Geometric Parameters

From the OXI geometric perspective:

- $\hbar_{i\alpha}$ appears as the "radius" of the action shell in body subspace
- $h_{i\alpha}$ is the "diameter" or scale for the timespace subspace

This treatment relates h and $\hbar$ through torus geometry:

- $h : T^1_{\text{torus}}(R) :: \hbar : T^2_{\text{torus}}(r, R)$

When we invoke timespace $\{t_i\}$, we need 3D h (for energy/time) and 3D $\hbar$ (for momentum/space), paired as $\{h_T, (h_{t_i}, \hbar_i)\}$.

To derive the geometric parameters (R_i, r_i) from Planck constants $(h, \hbar)$, we define:

$$h_{t_i} = \kappa_a r_i^2, \quad \hbar_i = \kappa_s R_i \tag{6}$$

With the relationship $R_i = r_i^2/f$, we establish:

$$\hbar_i = \kappa_s \frac{r_i^2}{f} = \frac{\kappa_s}{\kappa_a f} h_{t_i} = K_i h_{t_i} \tag{7}$$

Where $K_i = \kappa_s/(\kappa_a \cdot f)$ is a dimensionless constant relating the two action quanta.

6 Quantum-OXI Mapping for the Three-Body Problem

We now establish a formal mapping between the quantum formulation and our three-body OXI framework.

6.1 Fundamental Mappings

For each binary pair (i, j), we establish:

$$\hbar_{ij} = \kappa_s R_{ij} = \kappa_s a_{ij}(1 + e_{ij}) \tag{8}$$

$$h_{ij} = \kappa_a r_{ij}^2 = \kappa_a a_{ij}^2(1 - e_{ij}^2) \tag{9}$$

Where:

- a_{ij} is the semi-major axis of binary pair (i, j)

- e_{ij} is the orbital eccentricity
- κ_s and κ_a are scaling constants with dimensions [action/length] and [action/length^2] respectively

6.2 Tilt Angle Relationships

The OXI framework tilt angles map to quantum parameters:

$$\alpha_{ij} = \frac{1}{2}\tan^{-1}\left(\frac{2\beta_{ij}}{1-\beta_{ij}^2}\right) = \frac{1}{2}\tan^{-1}\left(\frac{2\sqrt{G(m_i+m_j)/a_{ij}}}{c - G(m_i+m_j)/a_{ij}}\right) \tag{10}$$

$$\delta_{ij} = \arctan(e_{ij}) = \arctan\left(\sqrt{1-\frac{r_{ij}^2}{R_{ij}^2}}\right) = \arctan\left(\sqrt{1-\frac{h_{ij}/\kappa_a}{(\hbar_{ij}/\kappa_s)^2}}\right) \tag{11}$$

This explicitly connects our tilt angles to the quantum geometric parameters.

6.3 Quantum State Representation

Each configuration of the three-body system is represented by quantum numbers:

$$|\Psi\rangle = |n_{12}, m_{t_{12}}\rangle \otimes |n_{23}, m_{t_{23}}\rangle \otimes |n_{31}, m_{t_{31}}\rangle \tag{12}$$

Where:

- n_{ij} represents spatial excitation (orbital angular momentum quantum number)
- $m_{t_{ij}}$ represents temporal excitation (radial quantum number)

The energy of each configuration is:

$$E = \sum_{ij\in\{12,23,31\}} \left[\frac{n_{ij}\hbar_{ij}^2}{2\mu_{ij}R_{ij}^2} + \frac{m_{t_{ij}}h_{ij}}{T_{ij}}\right] \tag{13}$$

Where:

- μ_{ij} is the reduced mass of pair (i, j)
- T_{ij} is the orbital period

6.4 Residue Accumulation as Quantum Transitions

The discrete residue updates correspond to quantum jumps:

$$\Delta n_{ij} = \left\lfloor \frac{\Delta L_{ij}}{\hbar_{ij}} \right\rfloor \tag{14}$$

$$\Delta m_{t_{ij}} = \left\lfloor \frac{\Delta E_{ij} T_{ij}}{h_{ij}} \right\rfloor \tag{15}$$

When sufficient action accumulates to change a quantum number, the system transitions to a new eigenstate.

These quantum jumps can also be represented using ladder operators:

$$\hat{a}_{ij} = \sqrt{\frac{1}{2\hbar_{ij}}} \cdot r_{ij} \tag{16}$$

$$\hat{p}_{ij} = \sqrt{\frac{\hbar_{ij}}{2}} \cdot \frac{\partial}{\partial r_{ij}} \tag{17}$$

With action states in Dirac notation:

$$|n_{ij}, m_{t_{ij}}\rangle = \frac{1}{\sqrt{n_{ij}! m_{t_{ij}}!}} (\hat{a}_{ij}^\dagger)^{n_{ij}} (\hat{b}_{t_{ij}}^\dagger)^{m_{t_{ij}}} |0, 0\rangle \tag{18}$$

In this representation, residue accumulation becomes:

$$\Delta R_{ij} \to \hat{a}_{ij}^\dagger |n_{ij}\rangle \tag{19}$$

$$\Delta E_{ij} \to \hat{b}_{t_{ij}}^\dagger |m_{t_{ij}}\rangle \tag{20}$$

6.5 Total Action and the R_7^{OV} Matrix

The total action of the system is:

$$h_T = \sum_{ij \in \{12,23,31\}} [\alpha_{ij} h_{ij} + \beta_{ij} \hbar_{ij}] \tag{21}$$

Where α_{ij} and β_{ij} are weighting factors that satisfy:

$$\alpha_{ij} + \beta_{ij} = 1 \tag{22}$$

In the three-body context, these weighting factors represent how energy is distributed between radial and angular components of orbital motion:

$$\alpha_{ij} = \frac{E_{\text{radial}}}{E_{\text{total}}} = \frac{1 - e_{ij}^2}{2 - e_{ij}^2} \tag{23}$$

$$\beta_{ij} = \frac{E_{\text{angular}}}{E_{\text{total}}} = \frac{1}{2 - e_{ij}^2} \tag{24}$$

The R_7^{OV} matrix elements become functions of these quantum parameters:

$$R_7^{OV}(ij,kl) = f\left(n_{ij}, m_{t_{ij}}, n_{kl}, m_{t_{kl}}, \frac{\hbar_{ij}}{\hbar_{kl}}, \frac{h_{ij}}{h_{kl}}\right) \tag{25}$$

Specifically:

- Diagonal elements represent self-evolution of each binary pair
- Off-diagonal elements represent coupling between different pairs
- The coupling strength depends on the ratio of quantum constants

6.6 Action Operator Formalism

We can define action operators that directly measure the accumulated residue:

$$\hat{J}_{ij} = \hbar_{ij}\left(n_{ij} + \frac{1}{2}\right) \tag{26}$$

$$\hat{H}_{t_{ij}} = h_{ij}\left(m_{t_{ij}} + \frac{1}{2}\right) \tag{27}$$

The total action operator becomes:

$$\hat{h}_T = \sum_{ij}\left[\alpha_{ij}\hat{H}_{t_{ij}} + \beta_{ij}\hat{J}_{ij}\right] \tag{28}$$

With expectation value:

$$\langle\hat{h}_T\rangle = \sum_{ij}\left[\alpha_{ij}h_{ij}\left(m_{t_{ij}} + \frac{1}{2}\right) + \beta_{ij}\hbar_{ij}\left(n_{ij} + \frac{1}{2}\right)\right] \tag{29}$$

This provides the total action of the three-body system in a given quantum state.

7 Real-Valued Evolution in R^7

For our R^7 framework, the evolution equation should remain real-valued:

$$\frac{\partial\Phi}{\partial\tau} = \hat{H}_7\Phi \tag{30}$$

Where $\hat{H}_7$ is a real differential operator in the 7D space. Complexity emerges only upon projection to lower dimensions.

7.1 Action-Angle Formalism

In the R^7 framework, we use real-valued operators:

$$\hat{a}_{ij} = \sqrt{\frac{1}{2\hbar_{ij}}} \cdot r_{ij} \tag{31}$$

$$\hat{p}_{ij} = \sqrt{\frac{\hbar_{ij}}{2}} \cdot \frac{\partial}{\partial r_{ij}} \tag{32}$$

The ladder operators emerge only upon projection to lower dimensions, where complex notation becomes useful.

7.2 Relationship Between Scaling Factors and Geometric Constants

The scaling factors κ_a and κ_s should be related by:

$$\kappa_s \cdot \kappa_a = \frac{G}{c^3} \tag{33}$$

Where G is the gravitational constant and c is the speed of light. This ensures compatibility with both quantum mechanics and general relativity in the appropriate limits.

We can also establish the relationship between our geometric parameters and quantum constants:

$$\frac{\hbar_{ij}}{h_{ij}} = \frac{\kappa_s}{\kappa_a} \frac{1 + e_{ij}}{a_{ij}(1 - e_{ij}^2)} \tag{34}$$

This ratio provides a direct measure of orbital geometry, connecting the quantum constants to the shape and size of the orbital ellipse.

7.3 Action Accumulation and Quantization

The quantization condition can be expressed as:

$$\oint p_{ij}\, dq_{ij} = n_{ij} \cdot \hbar_{ij} \tag{35}$$

With residue accumulation occurring when:

$$\int \Delta R_{ij}\, dt = \hbar_{ij} \tag{36}$$

This ensures proper accounting of action accumulation around closed orbits.

8 Transformation Properties and the R_7^{OV} Matrix

8.1 Quantum State Transformations

Under a change of reference frame, the quantum numbers transform as:

$$|n'_{ij}, m'_{t_{ij}}\rangle = \sum_{kl} D^j_{kl}(R_7^{OV})|n_{kl}, m_{t_{kl}}\rangle \tag{37}$$

Where D^j_{kl} are matrix elements of the representation of the rotation group, ensuring that our quantum state transforms correctly under vantage changes.

8.2 Exponential Generator Formulation

The R_7^{OV} matrix can be expressed through an exponential generator formulation:

$$R_7^{OV} = \exp\left(\sum_i \theta_i G_i\right) \tag{38}$$

Where:

- G_i are the generators of the SO(7) group
- θ_i are the corresponding rotation angles

This formulation provides the precise mathematical bridge between quantum state transitions and geometric rotations in 7D space. The generators G_i can be directly related to the action operators we've defined, creating a clean mapping between residue accumulation and matrix evolution.

8.3 R_7^{OV} Matrix and Quantum Operators

The R_7^{OV} matrix elements can be directly expressed in terms of quantum operators:

$$R_7^{OV}(ij, kl) = \langle n'_{ij}, m'_{t_{ij}}| \exp\Big(-i\theta_{ijkl}\hat{J}_{ijkl}/\hbar_T\Big)|n_{kl}, m_{t_{kl}}\rangle \tag{39}$$

Where $\hat{J}_{ijkl}$ is the appropriate angular momentum operator connecting the ij and kl sectors, and θ_{ijkl} is the rotation angle.

9 Characteristic Velocities

The characteristic velocities c_i can be related to the quantum constants through:

$$c_i = \sqrt{\frac{h_i \cdot \omega_i}{\hbar_i \cdot k_i}} = \sqrt{\frac{\kappa_a r_i^2 \cdot \omega_i}{\kappa_s R_i \cdot k_i}} \tag{40}$$

Where ω_i and k_i are the characteristic frequency and wavenumber, respectively. This explains why different characteristic velocities in different directions emerge naturally from the quantum geometry.

10 Discussion and Implications

Our quantum-OXI formulation of the three-body problem has several important implications that extend beyond celestial mechanics.

10.1 Relationship to Quantum Chaos

The discretized nature of our approach connects directly to quantum chaos theory. While classical chaos is characterized by exponential sensitivity to initial conditions, quantum chaos manifests differently due to the uncertainty principle. Our approach suggests that the three-body problem may exhibit features of both:

- Classical chaos in projected 3D trajectories
- Quantum-like predictability in the full 7D description with discrete transitions

This hybrid behavior may offer insights into the quantum-classical transition in chaotic systems.

10.2 Computational Advantages

The discretized approach potentially offers computational advantages over traditional integration methods:

- Instead of continuously tracking positions and velocities, we can propagate the system through discrete state transitions
- The quantum state representation may require fewer parameters than traditional phase space coordinates
- The formalism naturally preserves constants of motion through its quantum number representation

This could enable more efficient long-term simulations of gravitational systems.

10.3 Testable Predictions

Our framework makes several testable predictions:

- The existence of preferred configurations corresponding to eigenstates of the system
- Discrete jumps in orbital parameters when sufficient action accumulates
- Statistical patterns in the distribution of orbital parameters that reflect the underlying quantum structure

These could be tested with high-precision observations of triple star systems or experimental models.

11 Conclusion

We have developed a quantum-inspired formalism for the three-body problem based on the OXI framework. This approach interprets the three-body system as a "world piece computer" where each body performs its own physics computation by exchanging geometric information with other bodies.

The key insights of our approach include:

- Discrete residue updates that parallel quantum transitions
- A geometric interpretation of action quanta
- A mapping between orbital parameters and quantum numbers
- A formalism for representing the three-body state in terms of binary pair quantum numbers
- A mechanism for state transitions based on action accumulation

This formalism suggests that the apparent chaos in the three-body problem may emerge from projecting a well-ordered quantum system in 7D space down to observable 3D dynamics, similar to how relativistic effects in the OXI framework emerge from projections of higher-dimensional rotations.

Future work will focus on developing computational methods based on this formalism, exploring connections with quantum chaos theory, and investigating possible experimental tests of the discrete nature of three-body dynamics. Additionally, extending this approach to N-body systems and exploring its connections to quantum field theory may yield further insights into the fundamental nature of physical law.